1. The formula below can be used to describe a classical potential describing biological macromolecules. Briefly explain what each part of the formula represents.

$$U = \sum_{bonds} k_r(r - r_0)^2 + \sum_{angles} k_\theta(\theta - \theta_0)^2 + \sum_{dihedrals} C_k \cos(\varphi - \varphi_{0,k})) + \sum_{impropers} k_\alpha(\alpha - \alpha_0)^2 + \sum_{i<j} \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}} + \sum_{i<j} \left(\frac{A}{r_{ij}^{12}} - \frac{B}{r_{ij}^{6}}\right) + \sum_{h.b.} \left(\frac{C}{r_{ij}^{12}} - \frac{D}{r_{ij}^{10}}\right)$$

This expression is a classical force field potential energy written as a sum of bonded terms (bonds/angles/dihedrals/**Improper dihedrals**) and non-bonded terms (electrostatics/van der Waals/an explicit H-bond).

a) Bond stretching term

Deformation of chemical bond lengths away from their equilibrium value. "**Atoms oscillate around their equilibrium average distances; thus, chemical bonds are treated as harmonic oscillators.**"

r= bond length, r_0= equilibrium bond length, k_r= spring constant (bond stiffness).

b) Angle bending term

Deformation of bond angles away from equilibrium. Two consecutive bonds form an angle, we assume a **harmonic approximation**: the angle can oscillate around the equilibrium configuration with a well-defined angular strength.

θ= angle, θ_0= equilibrium angle, k_θ= angular stiffness.

c) Dihedral (torsional) term

The energetic preference for rotation about a bond, controlling conformations. "**Four consecutive atoms define three bonds and two planes. The two planes describe an angle which is called dihedral angle.**

φ: dihedral angle; C_k: strength for component k; $\varphi_{0,k}$: phase (preferred offset).

d) Improper dihedral term

Constraints that maintain planarity of specific atom groups. Carbon rings and other similar structure display planar configurations due to the specific electronic configurations of their atoms. Improper dihedral angles need to be introduced to keep such atoms in the same plane.

α= improper dihedral, α_0= target value, k_α= stiffness.

e) Coulomb (Electrostatic) non-bonded term

Coulomb interactions between atom pairs i, jthat are treated as non-bonded. The charge represents the average charge on the atom due to the distribution of the electrons in that particular molecule.

q_i, q_j: atomic charges; r_{ij}: distance between atoms $i$and j; $i < j$avoids double counting, ε_0: vacuum permittivity.

f) van der Waals (Lennard–Jones) term (non-bonded)

Non-bonded short-range repulsion (A/r^{12}) and longer-range **attraction** (dispersion) (B/r^6) between atoms $i$and j. It is the sum of an attractive part (quantum effect) and a repulsive part which represent the effect of electronic clouds in close proximity.

A, B: parameters; r_{ij}: interatomic distance; $i < j$avoids double counting.

g) Hydrogen-bond term

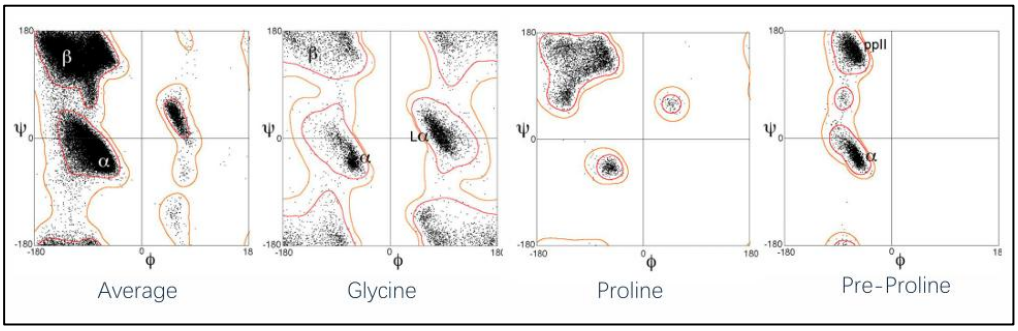
An empirical term to represent hydrogen-bond interactions. Hydrogen bonds: an empirical potential that take explicitly into account of the possibility of creation of hydrogen bonds.

C/rij^{12}: short-range repulsion; $-D/rij^{10}$: stabilizing attraction.

2. Figure shows the Ramachandran plot for four amino acids.

a) Identify which amino acid is proline and which is glycine and justify your answer.

b) Explain what the regions in the plot marked with α and β represent.



a) Average

This is the "typical" Ramachandran distribution for most residues: it shows the main allowed basins that correspond to common secondary-structure conformations.

The dense region labeled α corresponds to right-handed α-helix backbone geometry (approximately ϕ ≈ −60°, ψ ≈ −45°).

The dense region labeled β corresponds to extended β-strand / β-sheet-like backbone conformations (an extended region distinct from α).

b) Glycine

Compared with "Average", glycine shows a much broader allowed area (more ϕ/ψ combinations are populated).

Justification: glycine has the smallest side chain (H), so it has less steric hindrance, allowing more backbone conformations. This matches the lecture statement that different amino acids have different preferred orientation.

c) Proline

Proline shows a much more restricted distribution than "Average".

Justification: the proline side chain forms a ring with the backbone, which strongly constrains backbone geometry (especially ϕ), **particularly the ϕ angle**, so fewer (ϕ, ψ)combinations are sterically allowed.

d) Pre-Proline

This plot corresponds to residues immediately preceding a proline in sequence, and it shows a distinct preference pattern compared with "Average".

The labeled pPII region indicates a polyproline-II-type backbone conformation that is commonly enriched when the next residue is proline (the neighboring proline imposes steric constraints that shift the preferred ϕ/ψ space).

Because **ϕ and ψ are linked to the local secondary structure**, the labeled regions correspond to common backbone conformations: **α region**: backbone dihedral combinations typical of an **α-**

helix (right-handed). In the lecture table, α-helix angles are approximately **ϕ ≈ −60°** and **ψ ≈ −45°**.
β region: backbone dihedral combinations typical of **β-strands/β-sheets** (the β-sheet/strand area of the Ramachandran plot).

3. Figure shows part of a PDB file.

a) Explain what each of the column marked with a letter (A-J) represent.

b) Explain why different carbon atoms in column C are labeled differently.

ATOM	1	N	ARG	A	20	24.557	-38.438	-2.219	1.00	48.18	N
ATOM	2	CA	ARG	A	20	24.786	-38.383	-3.724	1.00	48.20	C
ATOM	3	C	ARG	A	20	26.209	-38.780	-4.233	1.00	46.86	C
ATOM	4	O	ARG	A	20	26.312	-39.492	-5.209	1.00	46.96	O
ATOM	5	CB	ARG	A	20	24.328	-37.035	-4.295	1.00	48.44	C
ATOM	6	CG	ARG	A	20	23.791	-37.076	-5.718	1.00	49.20	C
ATOM	7	CD	ARG	A	20	23.543	-35.655	-6.309	1.00	49.60	C
ATOM	8	NE	ARG	A	20	22.419	-34.974	-5.669	1.00	54.70	N
ATOM	9	CZ	ARG	A	20	22.453	-33.754	-5.107	1.00	55.47	C
ATOM	10	NH1	ARG	A	20	23.558	-33.001	-5.109	1.00	52.88	N
ATOM	11	NH2	ARG	A	20	21.345	-33.268	-4.550	1.00	55.28	N
ATOM	12	N	SER	A	21	27.281	-38.306	-3.583	1.00	45.68	N
ATOM	13	CA	SER	A	21	28.663	-38.666	-3.946	1.00	43.64	C
ATOM	14	C	SER	A	21	29.113	-39.741	-2.991	1.00	42.98	C
ATOM	15	O	SER	A	21	28.807	-39.631	-1.810	1.00	43.65	O
ATOM	16	CB	SER	A	21	29.618	-37.466	-3.846	1.00	43.49	C

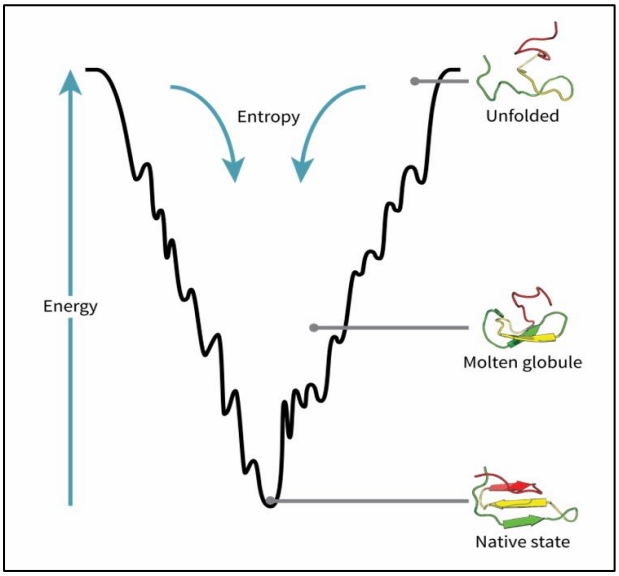
a) Everything following the 'ATOM' word indicates the properties of an atom in the protein.
Record type = ATOM (this line describes an atom record in a PDB file).
Atom number (atom serial index in the file).
Atom "Type"/Atom name (e.g., N, CA, C, O, CB, CG…).
Residue name (3-letter code).
Chain identifier (distinguishes different polymer chains).
Residue number (sequence index of the residue in that chain).
x coordinate (in Å).
y coordinate (in Å).
z coordinate (in Å).
Element (chemical element symbol, e.g., C, N, O).
Occupancy (fraction of time spent in that position for each atom).
B-factors (fluctuations around the average positions Å²).

b) Because a residue contains multiple distinct carbon atoms, the PDB format must assign unique atom names so each atom can be unambiguously identified in the structure.
In amino acids, the backbone carbons have standard names:
CA = α-carbon (the central carbon of the amino acid backbone)
C = carbonyl carbon of the peptide backbone
The side-chain carbons are named by position away from CA: CB, CG, CD …
Some residues have functional group-specific naming, e.g., ARG, because these are chemically different carbons in the side chain.
So different carbon atoms must be labeled differently

4. Figure above represents the "folding funnel" for a protein undergoing a folding process.

a) What is the relationship between the entropy of the protein and configurations it can assume? How does this entropy changes during the process?

b) What contributes to the internal energy of the protein (in term of molecular interactions), and how does the internal energy change during the process?



a) 1) Relationship: Conformational entropy reflects how many conformations (states) the protein can access. In the funnel model, "the width of the funnel represents conformational entropy (number of possible states), which decreases as folding progresses and the structure becomes more ordered.

Therefore: more accessible conformations → higher conformational entropy; fewer conformations → lower entropy.

2) During folding, the protein moves from an unfolded ensemble (many conformations) toward a more ordered native structure (fewer conformations). Thus, conformational entropy decreases as folding progresses and the structure becomes more ordered.

Multiple folding pathways can lead to the same native state, eliminating the need to search through all possible conformations randomly.

b) 1) the protein's internal energy (from interactions) is represented by a classical potential energy:

Bonded interactions: interactions that reproduce bond distances, bond angles, and related strengths. In typical force-field form this includes bond stretching, angle bending, dihedral (torsional) terms, and impropers.

Non-bonded interactions: Non-bonded atoms interact exclusively through Coulomb and Lennard-Jones potentials.

This corresponds to electrostatics (Coulomb) and van der Waals (Lennard–Jones) terms in U.

Hydrogen-bond potential (if included): the force-field example shown also includes an h.b. term

2) The funnel shape represents a bias toward the native state, with energy decreasing as the protein approaches its correctly folded structure.

Local minima in the funnel can act as folding intermediates or misfolded traps, but the overall downward slope drives folding toward the global energy minimum.

Folding is primarily driven by the hydrophobic effect, where hydrophobic residues collapse into the protein core to minimize unfavorable contacts with water, while hydrophilic residues remain on the surface.

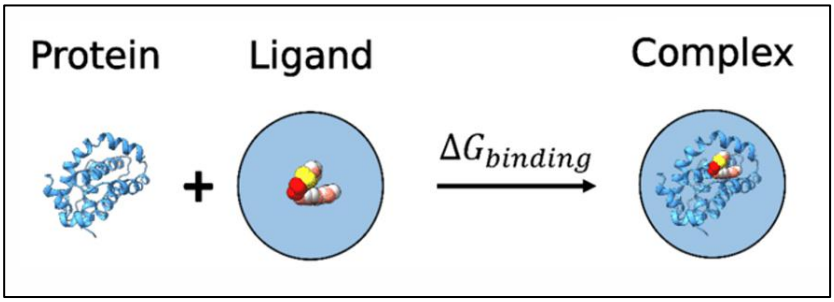
5. Figure schematically represents the binding process of a molecule to a protein.

a) The Gibbs Free Energy is given by: $G = U + pV - TS$. Explain what the internal energy $U$and

the entropy $S$are from the microscopic point of view.

b) Define the binding Free Energy and explain what the difference in free energies between the bound and unbound states represents thermodynamically. Briefly explain what a negative value for ΔGmeans.

c) Describe how molecular simulations can be used to evaluate internal energy and entropy contributions to binding free energy.



a) **Internal energy U** : microscopically, U is the system's energy of interaction between atoms/molecules, i.e., the (classical) potential energy describing how atoms interact.

Entropy S: microscopically, Sreflects the number of accessible microscopic states/configurations (configurational space).

b) **ΔG:** The binding free energy is the difference of free energy between the bound and unbound state.

$\Delta G_{binding}$=GB−GU

Thermodynamic meaning: The quantity $G_B - G_U$represents how much free energy changes upon forming the complex.

Negative ΔG: $\Delta G < 0$means binding is thermodynamically favourable, i.e., the bound state has lower free energy than the unbound state. This is consistent with the course link between **$\Delta G$and K_D: $\Delta G = RT\ln K_D$.**

c) **ΔG=GB−GU=UB−UU−T(SB−SU)**

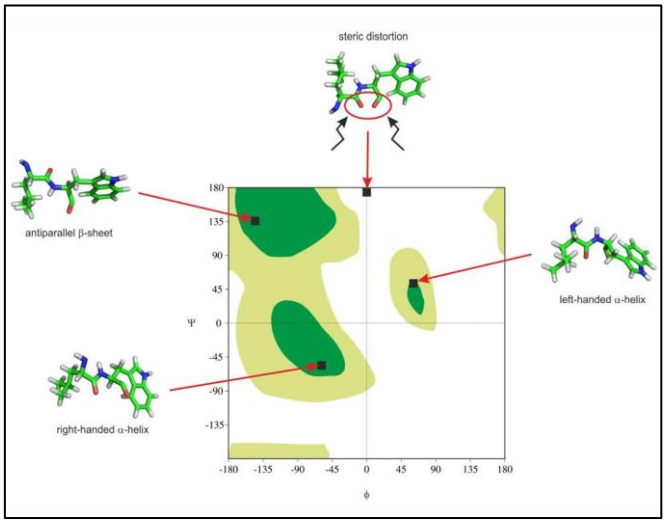
Estimate the internal energy contribution from the interaction energy of the system.

Estimate the entropy contribution from how the simulation samples the configurational space accessible to protein/ligand.

6. Figure shows the typical protein Ramachandran plots.

a) Provide a brief explanation of what the two angles (φ and ψ) in the plot are.

b) Write a short caption for Figure, explaining in particular what the light green and dark green regions represent.



a) The relative position of two consecutive amino acids in a polypeptide backbone is described by two dihedral (torsion) angles, ϕ and ψ.
ϕ (phi) is the backbone dihedral angle around the N–Cα bond.
ψ (psi) is the backbone dihedral angle around the Cα–C bond.
Not every (ϕ, ψ) combination is possible because of steric and electrostatic hindrance.
The values of ϕ and ψ are linked to the local secondary structure。

b) Ramachandran plot showing the distribution of backbone dihedral angles (ϕ, ψ) for amino-acid residues. The shaded regions represent (ϕ, ψ) combinations that are sterically/energetically allowed, while unshaded regions are disallowed due to steric and electrostatic hindrance. Dark green indicates the most favoured (most probable) conformational regions; light green indicates additional allowed but less favoured regions. Clusters correspond to common secondary structures such as the right-handed α-helix region (typical ϕ ≈ −60°, ψ ≈ −45°) and β-sheet regions.

7. **a) The binding free energy of a ligand (small molecule A) to a protein can be described by the formula ΔG=ΔH−TΔS. Explain what ΔG, ΔH and ΔS are in this context and how they can be interpreted at the microscopic level.**
b) Assume there is a second ligand (small molecule B) with binding affinity ΔGB<ΔGA. Explain how the two molecules will compete for binding to the protein if they are added to the solution at the same concentration.
c) Which of the two molecules in question (b) is a better drug candidate if we consider the binding affinity alone? Motivate your answer.

a) ΔG(binding free energy): the difference of free energy between the bound and unbound state,

$$\Delta G = G_B - G_U$$

and it can be written as

$$\Delta G = \Delta H - T\Delta S.$$

ΔH(enthalpy change upon binding): energy changes associated with breaking and forming intermolecular interactions (hydrogen bonds, van der Waals interactions, electrostatic interactions) during binding; it also includes (usually unfavorable) energy changes due to loss of solvation of the protein and ligand.

Microscopic interpretation: these are changes in the interaction (potential) energy between atoms/molecules.

ΔS(entropy change upon binding): change in the configurational space accessible to the proteins and ligands upon binding (usually unfavorable), plus release or reorganization of water molecules from the binding interfaces (can contribute favorably).

Microscopic interpretation: at physiological temperature multiple states/configurations are accessible ("Entropy exists"), and free energy determines the most probable state.

b) Comparing two ligands:

$\Delta\Delta G = \Delta G_B - \Delta G_A$. $\frac{K_{DA}}{K_{DB}} = e^{(\Delta G_A - \Delta G_B)/K_B T} = e^{-\Delta\Delta G/K_B T}$

If $\Delta G_B < \Delta G_A$, then $\Delta\Delta G < 0$, so B is a better binder than A (lower binding energy, better binding affinity).

Therefore, when A and B are added at the same concentration, B will preferentially occupy the protein binding sites (and will tend to displace A at equilibrium).

c) Ligand B, because $\Delta G_B < \Delta G_A$implies better binding affinity (better binder).

Equivalently: stronger binding corresponds to smaller K_D(and larger K_A, since $K_A = 1/K_D$).

8. Describe the importance of the –R group (side chain) in amino acid molecules.

The –R group is the variable part of an amino acid; it determines the amino acid's chemical/physical properties. Therefore, –R groups largely determine how amino acids interact, which in turn affects local conformational preferences and overall folding/structure. different amino acids have different preferred orientation (which explain their secondary structure propensity).

Are proteins with the same number of their respective amino acid components necessarily identical?

No. Proteins are defined by their primary structure, i.e., the linear sequence of amino acids linked by peptide bonds.

Two proteins can have the same length (same number of residues) but different amino-acid sequence and/or composition, leading to different structures and functions.

Even with the same length, different residues (different –R groups) give different interaction patterns and conformations.

Explain how protein folding is driven by the formation of a hydrophobic core.

Protein folding is driven by the hydrophobic effect, in which hydrophobic residues collapse into the protein core to minimize unfavorable contacts with water, while hydrophilic residues remain on the surface. This "hydrophobic collapse" produces a compact structure and promotes formation/stabilization of the native 3D fold.

Explain why the hydrophobic effect, rather than hydrogen bond formation, is the dominant factor that drives the folding of protein molecules.

Hydrogen bonds are not the dominant driving force because, in the unfolded state, many polar groups already form hydrogen bonds with water; during folding, many protein–water H-bonds are essentially exchanged for protein–protein H-bonds (often giving a smaller net driving

contribution).
By contrast, burying hydrophobic side chains reduces unfavorable contacts with water and is the primary driver of folding via the hydrophobic effect.

What are the two amino acids for which the Ramachandran plots differ significantly, and what is the reason for this difference?

Glycine and Proline.
Glycine: smallest side chain (R = H) → minimal steric hindrance → much larger allowed ϕ/ψ region.
Proline: cyclic side chain constrains the backbone → restricted ϕ angle → much smaller allowed ϕ/ψ region.
(General steric/constraint rationale is consistent with the course statement that not all ϕ/ψ combinations are allowed due to steric/electrostatic hindrance and that different amino acids have different preferred orientations.)
Alanine (Ala): side chain $–CH_3$
Glycine (Gly): side chain –H
Serine (Ser): side chain $–CH_2OH$

How does the Ramachandran diagram define restrictions on backbone conformation? Explain with examples.

Restrictions

The relative position of two consecutive amino acids is described by two backbone dihedral angles ϕ (phi) and ψ (psi).
Not every combination of ϕ and ψ is possible, because of steric and electrostatic hindrance.
Therefore, the Ramachandran diagram (plot of ϕ vs ψ) defines the allowed/forbidden backbone conformations by showing which (ϕ, ψ) combinations are feasible.
The values of these dihedral angles are linked to the local secondary structure.

Examples

α-helix region: typical dihedrals around ϕ ≈ −60° and ψ ≈ −45°.
The plot also contains regions corresponding to other secondary structures (e.g., β-sheet), consistent with "ϕ/ψ values are linked to local secondary structure".
Different amino acids have different preferred orientations (e.g., Glycine vs Proline families), which explains differences in secondary-structure propensity and their allowed regions.

Explain the concept of the protein folding funnel and its significance in protein structure formation

A possible way to solve Levinthal's paradox is assuming that the energy landscape is shaped as a funnel.
The funnel shape represents a bias toward the native state, with energy decreasing as the protein approaches its correctly folded structure.
Multiple folding pathways can lead to the same native state, eliminating the need to search through all possible conformations randomly.
The width of the funnel represents conformational entropy (number of possible states), which decreases as folding progresses and the structure becomes more ordered.
Local minima in the funnel can act as folding intermediates or misfolded traps, but the overall

downward slope drives folding toward the global energy minimum.

Folding is primarily driven by the hydrophobic effect, where hydrophobic residues collapse into the protein core to minimize unfavorable contacts with water, while hydrophilic residues remain on the surface.

Discuss the role of entropy in the folding process

The width of the funnel represents conformational entropy (number of possible states).

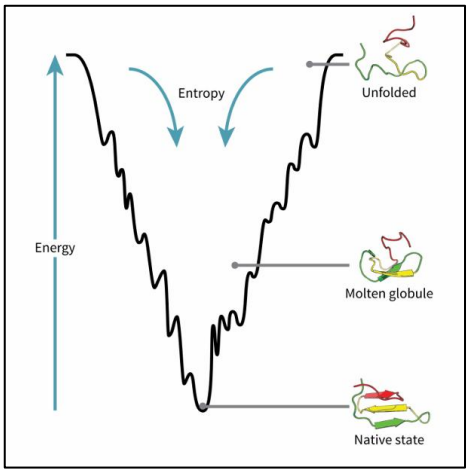
During folding, this width decreases, meaning conformational entropy decreases as folding progresses and the protein becomes more ordered.

Unfolded state: many accessible conformations → high conformational entropy (wide funnel).

Support your answer with a well-labelled diagram

Native state: far fewer conformations → low conformational entropy (narrow funnel).

Support your answer with a well-labelled diagram





Explain the concept of free energy landscapes in biomolecular systems, discuss their role in understanding conformational changes

Concept

A free energy landscape describes how the free energy of a biomolecular system changes as a function of its configuration/conformation. Different configurations can be accessible, and free energy determines the most probable state of the system.

Role in configurational changes

Conformational changes can be viewed as moving between states (minima) on this landscape by crossing free-energy barriers. The rates of transitions depend on the probability of "jumping" these barriers, which is exponential in the barrier free energy.

Why is this simple form of the diagram still an effective predictor of protein backbone

conformation?

The simplest form of the Ramachandran diagram is calculated by ignoring hydrogen bonding, interactions with water, and the hydrophobic effect.

Even the simplest Ramachandran diagram is effective because not every (ϕ, ψ) combination is allowed due to steric and electrostatic hindrance.

How do other forces such as hydrogen bonding and van der Waals interactions influence the accuracy of the diagram?

Hydrogen bonding and van der Waals interactions (and other interactions such as solvent/hydrophobic contributions) do not remove the sterically forbidden regions, but they change the relative stability within the allowed regions, making some (ϕ, ψ) zones more favorable/occupied than others (therefore sharpening/reshaping the "preferred" areas compared with the purely steric picture).

Explain the principle of Surface Plasmon Resonance (SPR) and how it works

SPR is an experimental method used to measure binding and unbinding of two molecules as a function of concentration.

In an SPR setup, one interaction partner is immobilised on a sensor surface, and the other is flowed over it. Binding at the surface produces a **time-dependent signal** ("sensorgram"), which reports the formation and loss of complex in real time.

Microscopically, binding is treated as a **reversible reaction**:

$$P + L \rightleftharpoons PL$$

with association and dissociation kinetics governed by:

$$\frac{d[PL]}{dt} = k_{\text{on}}[P][L] - k_{\text{off}}[PL]$$

Explain molecular polarization. Distinguish permanent vs induced dipole. Why is water a dipole but CO_2 is not? What happens in an external field?

How does the environment (solvent) change electrostatic interactions? What is a solvation shell and why does it matter?

Explain the physical meaning of the LJ terms (repulsive vs attractive). Why must energy be minimized before MD / after adding hydrogens / after docking?

Describe polymerization and pairing of nucleotides; relate to stability/structure. Compare DNA vs RNA structural roles.

What are ϕ and ψ? Why are not all combinations allowed? How do different residues (gly/pro) affect Ramachandran preferences?

What is homology modeling? State the basic assumption and why it can often give accurate predicted models.

What is ab initio (de novo) modeling? Why is it time-consuming, and what does it mean to "sample the conformational space"?

List and briefly explain the key innovations of AlphaFold.

State Coulomb's law and explain the meaning of each term (including the direction of the force). Define electrostatic potential $V(\mathbf{r})$and explain its relation to potential energy $U(\mathbf{r})$.

Explain the relation between electric field and electrostatic potential ($\mathbf{E} = -\nabla V$). What is an equipotential surface and how is it oriented with respect to field lines?

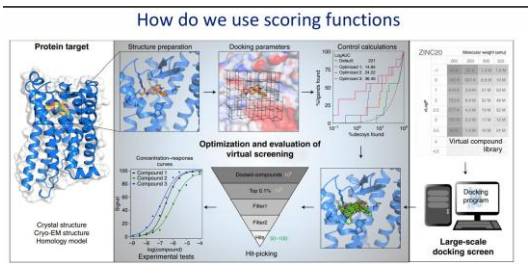
What is polarization of atoms/molecules? Explain how an external electric field can induce a dipole moment, and how this relates to partial charges in molecules.

Explain why screening is common in protein–ligand docking but usually not required in protein–protein docking.

Classify amino acids by side-chain properties; explain why special cases matter.
RNA primary structure is the linear sequence of nucleotides in the RNA chain。
RNA secondary structure refers to the local base-pairing pattern formed mainly by intra-molecular pairing within a single RNA strand.
RNA tertiary structure is the 3-D arrangement formed when multiple secondary-structure elements pack together through long-range interactions,

Cysteine contains a thiol side chain (–SH) that can form disulfide bridges (S–S) between two cysteine residues. Disulfide bridges are one of the interactions that stabilize the tertiary structure of proteins.

1. Molecular Docking:
   Autodock
   Autodock potential is based on Physics, but it introduce weighting pre-factor to each term (W) to better fit empirical data.
   Schoedinger and Glide
   Schoedinger is a software that use an empirical scoring function to determine the best mode and evaluate the binding energy of the docking.



2. Protein–Protein: HADDOCK, AF3 accurately predicts structures across biomolecular complexes

Molecular Docking

Protein-Ligand interactions

- Ligands are usually small and more rigid than proteins (few degrees of freedom).
- Small molecules usually bind to pockets (cavities).
- Interactions are often mediated by hydrophobic contributions.
- In many practical cases, a large number of ligands need to be tested (screened).

Protein-Protein interactions

- Proteins are large and can have important configuration changes upon binding.
- The interaction between the proteins happens on their surface.
- It is often mediated by hydrophilic residues and by surface complementarity.
- The interacting proteins are usually known (no need for screening).

Homology

Ab initio

Alphafold

Attention-based neural network evolutionary information end-to-end approach